

Model Selection: Part one

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Wednesday 19th November, 2025

Applied Econometrics

AIC/BIC and KLIC

How many components should we include in our model?

- ▶ Too few: under-fitting and large residuals.
- ▶ Too many: over-fitting and poor out of sample prediction.

How do we choose?

- ▶ X variables.
- ▶ Instrumental Variables.

When do we have too much data?

- ▶ On the internet!
- ▶ Hedonics: What really determines the price of your house?
- ▶ Prediction: What really determines loan defaults?
- ▶ Consideration Sets: How many products do consumers really choose among on the shelf?
- ▶ Which elements of financial filings really matter?

What do people mostly do in practice?

- ▶ Regress Y on X with all variables included.
- ▶ Drop some variables if they aren't significant?
- ▶ Re-run with some things dropped
- ▶ Add in some other things that may or may not be significant.

Nested and Non-nested Models

What makes a model **nested** or **non-nested**?

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 w_i + \beta_3 z_i + \varepsilon_i$$

A nested model can be written as a restricted version of the larger model

- ie: all of the following are nested within the model above

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 w_i + \varepsilon_i$$

$$y_i = \beta_0 + \beta_1 x_i + \beta_3 z_i + \varepsilon_i$$

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- ie: this model is non-nested (because of s_i)

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 w_i + \beta_3 z_i + \gamma s_i + \varepsilon_i$$

What we teach undergrads?

Start with sum of squared errors (If you want $\frac{1}{n}$'s imagine them):

$$\underbrace{\sum_{i=1}^n (y_i - \bar{y})^2}_{\text{total sum of squares}} = \underbrace{\sum_{i=1}^n (y_i - \hat{y}_i(\theta))^2}_{\text{residual sum of squares}} + \underbrace{\sum_{i=1}^n (\hat{y}_i(\theta) - \bar{y})^2}_{\text{explained sum of squares}}$$

Let $\dim(\theta) = p$ (the number of parameters).

Three traditional ways to select the number of components in a model:

$$\overline{R}^2 = 1 - SSR(p)/TSS - SSR(p)/TSS \cdot \frac{p}{N - p - 1}$$

$$AIC(p) = \ln \left(\frac{SSR(p)}{N} \right) + (p + 1) \frac{2}{N}$$

$$BIC(p) = \ln \left(\frac{SSR(p)}{N} \right) + (p + 1) \frac{\ln N}{N}$$

These are designed for strictly **nested** models.

- ▶ AIC tends to select larger models than BIC since it penalizes the number of parameters less heavily.
- ▶ These usually depend on ordering potential models by p the number of components and then sequentially fitting them.
- ▶ AIC is not consistent: as $N \rightarrow \infty$ it may still select too many parameters.
- ▶ BIC is consistent: as $N \rightarrow \infty$ it will select the correct number of parameters.
- ▶ Of course for finite-sample $N < \infty$ anything can happen.

What is KLIC?

Kullback-Leibler information criterion:

$$\begin{aligned} KLIC(f, g) &= \int \mathbf{f}(\mathbf{y}) \log \left(\frac{\mathbf{f}(\mathbf{y})}{\mathbf{g}(\mathbf{y})} \right) d\mathbf{y} \\ &= \int f(y) \log(f(y)) \partial y - \int f(y) \log(g(y)) \partial y \\ &= C_f - \mathbb{E}_f \log(g(y)) \end{aligned}$$

Observe $KLIC(f, g) \geq 0$ and $KLIC(f, g) = 0$ IFF f, g are the same distribution!

C_f we ignore (doesn't depend on g).

Where does it come from?

How do we come up with these penalized regressions?

- ▶ AIC/BIC arise from considering the likelihood ratio test (LRT) of a maximum likelihood estimator and making a lot of assumptions.
- ▶ AIC arises from minimizing the Expected KLIC.
- ▶ Picking a model with best AIC means picking a model based on (estimated) expected KLIC (if g includes the correct model).
- ▶ Low values of KLIC mean the models are similar.

Where does it come from?

How do we come up with these penalized regressions?

- Recall that OLS is a ML estimator in the case where ε is normally distributed.

$$D = -2 \ln \left(\frac{\text{Likelihood } H_0}{\text{Likelihood } H_a} \right) = -2 \ln \left(\underbrace{\frac{(\sup L(\theta|x) : \theta \in \Theta_0)}{(\sup L(\theta|x) : \theta \in \Theta)}}_{\Lambda(x)} \right)$$

- If the models are **nested** then $\Theta_0 \subset \Theta$ and $\dim(\Theta) - \dim(\Theta_0) = q$ then as $N \rightarrow \infty$ we have that $D \rightarrow^d \chi^2(q)$.

Many cases we are interested in are **not strictly nested**

- ▶ Should I include x_2 OR x_3 in my regression? (partially overlapping)
- ▶ Is the correct distribution $f(y|x, \theta)$ normal or log-normal? (non-overlapping)

- ▶ Cox (1961) suggested the following (often infeasible solution) by assuming that F_θ is the true model.

$$LR(\hat{\theta}, \hat{\gamma}) = L_f(\hat{\theta}) - L_g(\hat{\gamma}) = \sum_{i=1}^N \ln \frac{f(y_i|x_i, \hat{\theta})}{g(y_i|x_i, \hat{\gamma})}$$

- ▶ Depending on which the true model is you could reject F_θ for G_γ and vice versa!
- ▶ Deriving the test statistic is hard (and specific to F_θ) because we must obtain $\mathbb{E}_f[\ln \frac{f(y_i|x_i, \hat{\theta})}{g(y_i|x_i, \hat{\gamma})}]$.
- ▶ Similar to AIC in that we are minimizing KLIC over F_θ .

$$\begin{aligned} H_0 : \mathbb{E}_{h(y|x)} \left[\frac{f(y|x, \theta)}{g(y|x, \gamma)} \right] &= 0 \\ \rightarrow \mathbb{E}_h[\ln(h/g)] - \mathbb{E}_h[\ln(h/f)] &= 0 \end{aligned}$$

- ▶ Instead of taking expectation with respect to one of two distributions, we take it with respect to $h(y|x)$ the unknown but **true distribution**.
- ▶ Same as testing whether two densities (f, g) have same KLIC.
- ▶ The main result is that (details in 8.5 of CT):

$$\frac{1}{\sqrt{N}} LR(\hat{\theta}, \hat{\gamma}) \rightarrow^d N[0, \omega_*^2], \quad \omega_*^2 = V_0 \left[\ln \frac{f(y|x, \hat{\theta})}{g(y|x, \hat{\gamma})} \right]$$

- ▶ Model selection is not the same thing as significance of β .
- ▶ AIC/BIC (even $\overline{R}^2$) compare models based on goodness of fit.
- ▶ BIC selects model on highest posterior probability of being the true model.
- ▶ AIC selects model that minimizes expected KLIC to the data.
- ▶ In practice both assume something like a likelihood and construct a penalty term.